

TANUSH THAKRAN

AI / ML Engineer · Applied AI Developer · Full-Stack (MERN · Flutter)

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[Portfolio](#) · [LinkedIn](#) · [GitHub](#)

PROFILE

AI-focused developer building applied machine learning and LLM systems, from retrieval-augmented generation pipelines to NLP-driven products. Currently pursuing an Integrated B.Sc. in Computer Science at Scaler School of Technology (in collaboration with BITS Pilani). Comfortable across the full stack — Python for AI/data workloads and MERN + Flutter for shipping intelligent, user-facing applications. Seeking AI/ML engineering and applied-AI roles where I can turn models into production systems.

TECHNICAL SKILLS

AI / ML: Retrieval-Augmented Generation (RAG) · LLM Integration · Prompt Engineering · NLP · Embeddings & Vector Search · Whisper (Speech-to-Text) · Ollama (Local LLMs) · Cosine Similarity Retrieval

ML & Data Tools: NumPy · Pandas · scikit-learn (data workflows) · joblib · deepseek-r1 · bge-m3

Languages: Python · JavaScript (ES6+) · Java · Dart · HTML5 / CSS3

Frameworks: React.js · Node.js · Express.js · Tailwind CSS · Flutter

Data & Infra: MongoDB · Firebase · REST APIs · Git · GitHub

Dev Tools: Cursor · GitHub Copilot · Windsurf · Vercel · Netlify · Render

AI / ML PROJECTS

RAG-Based AI — Offline Retrieval-Augmented QA Pipeline

Python · Ollama · Whisper

github.com/Tanush1206/rag_based_ai

- Built a fully offline RAG pipeline that ingests course videos and answers natural-language questions grounded strictly in retrieved transcript context — no external APIs, everything runs locally.
- Engineered the end-to-end flow: audio extraction (ffmpeg), timestamped transcription with Whisper, embedding generation via the local bge-m3 model, and top-k retrieval using cosine similarity.
- Fed only the most relevant chunks to a local deepseek-r1 LLM through Ollama, with prompts constraining answers to cited video, title, and timestamp for traceable, hallucination-resistant responses.
- Optimized embedding throughput by batching chunk requests to avoid runner crashes on large transcript sets.

PYTHON · RAG · WHISPER · OLLAMA · EMBEDDINGS · VECTOR RETRIEVAL · LOCAL LLM

PactPal — AI Legal Agent

Python · NLP · React

pactpal-frontend.onrender.com

- AI-powered platform that converts complex legal contracts into plain language using NLP, reducing reading time by ~30%.
- Designed for non-legal users — startups and individuals — enabling faster, informed decisions without legal expertise.

PYTHON · NLP · REACT · PRODUCTION

ADDITIONAL PROJECTS

ATW Mobile — Driver & Medic Coordination App

Flutter · Firebase

github.com/NeuralSynth/cyparta-atw-frontend-android

- High-criticality logistics app with real-time GPS and full trip-lifecycle management; shipped as a production intern project for CYPARTA, Cairo with end-to-end ownership.

FLUTTER · FIREBASE · GPS TRACKING · REAL-TIME

CLIQUE — Event Platform

React · Node.js

frontend-clique-1.onrender.com

- Event tracking and centralization platform helping Scaler students manage academic and extracurricular schedules.

REACT · NODE.JS · LIVE

AASRAH — Humanitarian Platform

React · Node/Express

github.com/NoiceHax/Aasrah

- Connects users with NGOs for real-time assistance during human or animal distress, with Google Maps API location tracking for rapid emergency response.

REACT · GOOGLE MAPS API · NODE / EXPRESS

EXPERIENCE

App Developer Intern — CYPARTA

Cairo, Egypt · Dec 2025 – Jan 2026

- Developed the Driver & Medic mobile application focused on high-criticality logistics and real-time coordination.
- Implemented end-to-end trip workflows: request acceptance, live status tracking, and navigation integration.
- Engineered post-trip EMS reporting modules for data integrity and rapid documentation.
- Collaborated on real-time GPS tracking and in-app notifications for seamless driver–medic communication.

FLUTTER · FIREBASE · GPS · API INTEGRATION · REAL-TIME

Junior Manager — AIESEC in Bengaluru

Oct 2025 – Mar 2026

- Managed operations for the Outgoing Global Volunteer department, facilitating international leadership programs.
- Transitioned to Incoming Global Volunteer, overseeing project management and intern integration.
- Led cross-functional teams to hit departmental KPIs and expand organizational reach across Bengaluru.

LEADERSHIP · TEAM MANAGEMENT · OPERATIONS · GLOBAL EXCHANGE

EDUCATION

Scaler School of Technology

2024 – 2028 · Bengaluru, India

Integrated B.Sc. in Computer Science (in collaboration with BITS Pilani) — Focus: Algorithms, System Design, Software Engineering Practices.

Columbia Foundation School

2022 – 2024

Class 11th & 12th — Science & Mathematics.

CERTIFICATIONS

- AI Fluency: Framework & Foundations — Anthropic (Jul 2026)
- Intro to Machine Learning — Kaggle (Jul 2026)
- The Ultimate Job Ready Data Science Course — CodeWithHarry
- Python Course — Tutedude (Apr 2026)
- Learn JavaScript — Scrimba (Feb 2026)